Decision Models in GeoAI:

Testing Jev for qualifying spatial relations.

Master's Thesis

Jakob Roland

October 26, 2026

Abstract

Contents

Abstract **1**

1 Introduction **2**

2 Research Goals **3**

3 Related Works **4**
3.1 GeoAI 4
3.2 General versus Decisional Intelligence 4
3.3 Benchmarking Jev 4
3.4 Ethical Remarks to GeoAI 4
3.5 Qualifying Spatial Relations 4

4 Experiment **5**
4.1 Data 5
4.2 Prompts 5
4.3 Results 5

5 Discussion **6**

6 Conclusion **7**

Bibliography **8**

1. Introduction

2. Research Goals

3. Related Works

3.1 GeoAI

3.2 General versus Decisional Intelligence

3.3 Benchmarking Jev

3.4 Ethical Remarks to GeoAI

3.5 Qualifying Spatial Relations

4. Experiment

4.1 Data

4.2 Prompts

4.3 Results

5. Discussion

6. Conclusion

Bibliography

[1] T. Deußer, L. Sparrenberg, and R. Sifa. “Evaluating and Benchmarking the System One Model Jev.” arXiv: 2609.37647 [cs.CL], Accessed: Oct. 26, 2026. [Online]. Available: http://arxiv.org/abs/2609.37647 pre-published.

Appendix

Use of Artificial Intelligence

For this thesis, Anthropic Claude Opus 5.5 [66] was employed to assist in refining and improving the text throughout all sections of this paper. This model was also used to help write the code for this paper, which is available at github.com/AppliedMachineLearn Lab/jev-benchmarking. The authors retain full responsibility for the accuracy, integrity, and originality of the work. (Template from [1])